

Exact Invertibility Probability for Binary Matrices with Two Ones in Every Row and Column

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Abstract

Let A_N be uniform among the $N \times N$ zero-one matrices having two ones in every row and every column, with diagonal ones allowed. We prove an exact coefficient formula for $\mathbb{P}(\det A_N \neq 0)$ for every $N \geq 2$. If

$$D(z) = e^{-z/2}(1-z)^{-1/2}, \quad H(z) = e^{-z/2}(1+z)^{1/4}(1-z)^{-1/4},$$

where the branches are defined by their power series at zero, then the probability is $[z^N]H(z)/[z^N]D(z)$. We also obtain the parity-sensitive expansion

$$\mathbb{P}(\det A_N \neq 0) = \frac{2^{1/4}\sqrt{\pi}}{\Gamma(1/4)}N^{-1/4} + (-1)^{N+1}\frac{e\sqrt{\pi}}{2^{9/4}\Gamma(3/4)}(N^{-3/4} + N^{-7/4}) + O(N^{-9/4}).$$

The proof derives the conditioned Ewens representation from its matching fibres, proves the cycle determinant criterion, and retains the contribution of the second dominant singularity at $z = -1$.

1 Introduction

For $N \geq 2$, set

$$\mathcal{M}_{N,2} = \{A \in \{0,1\}^{N \times N} : A\mathbf{1} = \mathbf{21}, A^T\mathbf{1} = \mathbf{21}\}.$$

Thus diagonal entries equal to one are permitted; equivalently, loops are allowed in the directed-graph interpretation. We study the exact probability that a uniform member of $\mathcal{M}_{N,2}$ is invertible over $\mathbb{R}$.

He and Huang represent this same loops-allowed model by a uniform permutation matrix and an independent Ewens permutation of parameter $1/2$, conditioned to have no fixed points [2, Lemma 6.2]. The determinant criterion in terms of the relative permutation's cycle lengths is already contained in the determinant classification of Li, Lin, and Rodman [4, Theorem 3.1]; earlier existence results are due to Houck and Paul [3]. We reproduce the structural arguments needed here so that the enumeration is self-contained.

The contribution is the resulting all- N probability formula and its explicit two-singularity asymptotic. The analytic argument is a direct Darboux calculation in the classical spirit of [1]. A distinct recent study by Zeng concerns the probability that a uniform permutation has pairwise distinct cycle lengths [5]; that event and probability law differ from those used below. We make no priority claim beyond this bounded comparison with the cited literature.

Theorem 1. Let A_N be uniform on $\mathcal{M}_{N,2}$ and put

$$d_N = [z^N]D(z), \quad h_N = [z^N]H(z),$$

where

$$D(z) = e^{-z/2}(1-z)^{-1/2}, \quad H(z) = e^{-z/2}(1+z)^{1/4}(1-z)^{-1/4}.$$

The fractional powers are the branches represented by their formal series at zero. For every $N \geq 2$, $d_N > 0$ and

$$\mathbb{P}(\det A_N \neq 0) = \frac{h_N}{d_N}. \quad (1)$$

More precisely,

$$|\mathcal{M}_{N,2}| = (N!)^2 d_N, \quad |\{A \in \mathcal{M}_{N,2} : \det A \neq 0\}| = (N!)^2 h_N. \quad (2)$$

The exceptional values are

$$p_2 = p_4 = 0, \quad p_3 = 1,$$

and $0 < p_N < 1$ for every $N \geq 5$. As $N \rightarrow \infty$ through all integers,

$$p_N = C_0 N^{-1/4} + (-1)^{N+1} C_1 (N^{-3/4} + N^{-7/4}) + O(N^{-9/4}), \quad (3)$$

where

$$C_0 = \frac{2^{1/4} \sqrt{\pi}}{\Gamma(1/4)}, \quad C_1 = \frac{e \sqrt{\pi}}{2^{9/4} \Gamma(3/4)}.$$

In particular, $p_N \sim C_0 N^{-1/4}$.

2 Matching fibres and the relative permutation

For a permutation $q \in S_N$, write $c_k(q)$ for its number of k -cycles and $c(q) = \sum_k c_k(q)$. Let

$$\mathcal{D}_N = \{q \in S_N : c_1(q) = 0\}.$$

Proposition 2. Choose a permutation matrix P uniformly. Independently choose $Q \in \mathcal{D}_N$ with probability proportional to $2^{-c(Q)}$. Then $P(I + Q)$ is uniform on $\mathcal{M}_{N,2}$.

Proof. Associate $A \in \mathcal{M}_{N,2}$ with its bipartite graph on labelled row and column vertices. This is a simple 2-regular bipartite graph, hence a disjoint union of even cycles of lengths $2k$ with $k \geq 2$. If $c(A)$ denotes the number of components, each component has exactly two alternating perfect matchings. Consequently A has exactly $2^{c(A)}$ ordered decompositions

$$A = P + R$$

into disjoint permutation matrices.

For such a decomposition let $Q = P^{-1}R$, so $A = P(I + Q)$. Disjointness is equivalent to Q having no fixed point. Alternately following an R -edge and the inverse of a P -edge around a component of length $2k$ gives one k -cycle of Q . Thus $c(Q) = c(A)$.

Set $Z_N = \sum_{q \in \mathcal{D}_N} 2^{-c(q)}$. This is positive because an N -cycle belongs to $\mathcal{D}_N$. Under the law in the proposition, a fixed A has mass

$$\sum_{P(I+Q)=A} \frac{2^{-c(Q)}}{N! Z_N} = \frac{2^{c(A)} 2^{-c(A)}}{N! Z_N} = \frac{1}{N! Z_N},$$

which is independent of A . □

The weighting in Proposition 2 is precisely the Ewens $_N(1/2)$ law conditioned on having no fixed points. The proof also establishes its normalization in this application, rather than assuming the coefficient identity that will follow.

3 Determinants and exact enumeration

Proposition 3. *For $A = P(I + Q)$ as above, A is invertible if and only if every cycle of Q has odd length. Under the fixed-point-free condition, the allowed lengths are therefore $3, 5, 7, \dots$*

Proof. After a change of basis, the permutation matrix Q is block diagonal with one cyclic permutation block C_k for every k -cycle. Since the characteristic polynomial of C_k is $x^k - 1$,

$$\det(I + C_k) = 1 - (-1)^k.$$

The matrix P is invertible, so multiplicativity of the determinant gives the assertion. In fact, on the invertible event, $|\det A| = 2^{c(Q)}$. $\square$

We next enumerate the weighted relative permutations. For a set L of allowed cycle lengths, the standard cycle count gives

$$\begin{aligned} \frac{1}{N!} \sum_{\substack{q \in S_N \\ \text{all cycle lengths in } L}} 2^{-c(q)} &= [z^N] \prod_{k \in L} \sum_{m \geq 0} \frac{(z^k/(2k))^m}{m!} \\ &= [z^N] \exp\left(\frac{1}{2} \sum_{k \in L} \frac{z^k}{k}\right). \end{aligned} \quad (4)$$

Indeed, permutations with multiplicities (m_k) number $N!/\prod_k k^{m_k} m_k!$.

Taking $L = \{2, 3, \dots\}$ in (4) gives

$$\exp\left(\frac{1}{2} \sum_{k \geq 2} \frac{z^k}{k}\right) = e^{-z/2} (1 - z)^{-1/2} = D(z).$$

Taking $L = \{3, 5, 7, \dots\}$ and using $\sum_{k \geq 1, k \text{ odd}} z^k/k = \frac{1}{2} \log((1+z)/(1-z))$ gives $H(z)$. Propositions 2 and 3 now yield (1). Moreover $Z_N = N!d_N$, so the constant mass found in Proposition 2 proves both identities in (2).

The coefficients can be evaluated exactly without expanding fractional powers. Differentiation gives

$$2(1-z)D'(z) = zD(z), \quad 2(1-z^2)H'(z) = z^2H(z),$$

and hence

$$2nd_n = 2(n-1)d_{n-1} + d_{n-2} \quad (n \geq 2), \quad (5)$$

$$2nh_n = 2(n-2)h_{n-2} + h_{n-3} \quad (n \geq 3), \quad (6)$$

with $d_0 = h_0 = 1$, $d_1 = 0$, and $h_1 = h_2 = 0$. For illustration,

N	2	3	4	5	6	7	8	9
p_N	0	1	0	$\frac{12}{17}$	$\frac{16}{151}$	$\frac{480}{823}$	$\frac{1536}{10631}$	$\frac{8176}{15871}$

It remains to justify the qualitative assertions about small sizes. No sum of odd integers at least three equals 2 or 4, whereas 3 itself is allowed. Thus $p_2 = p_4 = 0$ and $p_3 = 1$. For odd $N \geq 5$, one N -cycle is an invertible witness; for even $N \geq 6$, cycles of lengths 3 and $N-3$ are an invertible witness. A single even N -cycle is a singular witness for even N , while cycles of lengths 2 and $N-2$ give one for odd $N \geq 5$. Every witness has positive weight, proving $0 < p_N < 1$ for $N \geq 5$. The same N -cycle argument gives $d_N > 0$ for all $N \geq 2$.

4 The two-singularity asymptotic

We use the following elementary coefficient estimate. For a fixed real nonintegral β , the generalized binomial identity and the first gamma-ratio term from Stirling's formula give

$$[z^n](1-z)^\beta = \frac{\Gamma(n-\beta)}{\Gamma(-\beta)\Gamma(n+1)} = \frac{n^{-\beta-1}}{\Gamma(-\beta)} \left(1 + \frac{\beta(\beta+1)}{2n} + O(n^{-2}) \right). \quad (7)$$

Replacing z by $-z$ contributes a factor $(-1)^n$.

For completeness, we record why the contributions at 1 and -1 may be added with the error terms used below. Expand each analytic multiplier at each of those points through degree three and subtract the resulting four fractional-power terms globally. For D , the remainder has local order $(1-z)^{7/2}$ at 1. For H , the local orders are $(1-z)^{15/4}$ at 1 and $(1+z)^{17/4}$ at -1 . The remainders therefore extend with three continuous derivatives to the unit circle. Three integrations by parts in the boundary Fourier coefficient formula show that their coefficients are $O(n^{-3})$. Applying (7) separately to the omitted quadratic and cubic terms shows that retaining only the constant and linear multiplier terms has error $O(n^{-5/2})$ for D and $O(n^{-11/4})$ for H . There are no other singularities on the unit circle.

Put $t = 1 - z$ and $s = 1 + z$. Direct Taylor expansion gives

$$D(z) = e^{-1/2} t^{-1/2} (1 + t/2 + O(t^2)), \quad (8)$$

$$H(z) = e^{-1/2} 2^{1/4} t^{-1/4} (1 + 3t/8 + O(t^2)) \quad (z \rightarrow 1), \quad (9)$$

$$H(z) = e^{1/2} 2^{-1/4} s^{1/4} (1 - 3s/8 + O(s^2)) \quad (z \rightarrow -1). \quad (10)$$

Define

$$a = \frac{e^{-1/2} 2^{1/4}}{\Gamma(1/4)}, \quad b = \frac{e^{1/2} 2^{-1/4}}{\Gamma(-1/4)}, \quad c = \frac{e^{-1/2}}{\sqrt{\pi}}.$$

Combining (7) with (8)–(10) yields

$$d_N = c N^{-1/2} \left(1 - \frac{3}{8N} + O(N^{-2}) \right) \quad (11)$$

and

$$h_N = a N^{-3/4} \left(1 - \frac{3}{8N} \right) + (-1)^N b N^{-5/4} \left(1 + \frac{5}{8N} \right) + O(N^{-11/4}). \quad (12)$$

For reference, the three relative N^{-1} corrections are

$$-\frac{1}{8} + \frac{1}{2} \left(-\frac{1}{2} \right) = -\frac{3}{8}, \quad -\frac{3}{32} + \frac{3}{8} \left(-\frac{3}{4} \right) = -\frac{3}{8}, \quad \frac{5}{32} - \frac{3}{8} \left(-\frac{5}{4} \right) = \frac{5}{8}.$$

The first entry belongs to D at 1, the second to H at 1, and the third to H at -1 .

Since $d_N > 0$, we may divide (12) by (11). The two nonoscillating corrections $-3/(8N)$ cancel. For the oscillating term, multiplication by the inverse denominator changes $5/8$ to $5/8 + 3/8 = 1$. Finally,

$$\frac{a}{c} = \frac{2^{1/4} \sqrt{\pi}}{\Gamma(1/4)}, \quad \frac{b}{c} = \frac{e \sqrt{\pi} 2^{-1/4}}{\Gamma(-1/4)} = -\frac{e \sqrt{\pi}}{2^{9/4} \Gamma(3/4)},$$

where $\Gamma(-1/4) = -4\Gamma(3/4)$. The numerator error in (12), after division, is $O(N^{-9/4})$; the relative $O(N^{-2})$ error in (11) contributes the same order to the leading term. This proves (3) and completes the proof of Theorem 1.

Computational and AI assistance

Deterministic Python scripts were used to enumerate all permutations for $2 \leq N \leq 9$, all matrices for $2 \leq N \leq 5$, and to check the coefficient recurrences and local multiplier algebra exactly. Floating-point evaluations were used only as diagnostics and not as proof. An AI language model assisted with derivation, algebraic cross-checking, and manuscript preparation. The mathematical argument needed for the theorem is given in full above; neither the finite computations nor the AI system is used as a substitute for proof.

References

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